

Karl Anthony Parks

I build rockets, robots, & programs.

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SUMMARY OF QUALIFICATIONS

- Passion for developing and debugging avionics systems for space exploration and innovation.
- Professional experience building complex mechanical, electrical, and software systems for the real world.
- Mechanical: CAD, FEA, and mesh modeling experience with PTC Creo/Simulate, SolidWorks, and Blender.
- Electrical: PCBs/KiCad, soldering, digital multimeters, and oscilloscopes.
- Software: MATLAB, Python, C/C++ (Bazel), TypeScript/JavaScript + React/Express, Kotlin/Java (Gradle), and LabVIEW.
- Fluids: LOX, LNG, N2O, HTP, LN2, Nitrogen, Helium, and pneumatic plumbing systems.
- Communications: gRPC, HTTP, WebSocket, UDP/TCP, CAN bus, Modbus, Ethernet, Wireless, LoRa, IPv4, Wireshark.

EDUCATION

San Diego State University (SDSU), CA

December 2019

Bachelor of Science in Aerospace Engineering, Minor in Computer Science

GPA: 3.5/4.0

RELEVANT EXPERIENCE

Parabilis Space Technologies, San Marcos CA (Jun 2018 - Feb 2020, Jul 2024 - Present)

(Designs, Develops, Tests, and Flies Affordable Propulsion, Launch Vehicles, and Spacecraft/Small Satellite Solutions)

- Aerospace Engineer (Jul 2024 - Present)
 - Developed and tested high-impulse, high-delta-v chemical propulsion hardware for a 6U CubeSat.
 - Led mechanical design of high test peroxide catalyst bed and acrylic motor.
 - Implemented new software tools that improved speed and quality of test preparation and execution.
- Engineering Intern (Jun 2018 - Feb 2020)
 - Designed and constructed a new hybrid rocket motor test stand in less than 3 months.
 - Developed a National Instruments-based DAQ system for strain gauge measurement and data logging.
 - Developed custom software for low-cost infrared thermal imaging FLIR camera for hot fire testing.

Boston Dynamics, Waltham MA (Mar 2020 - Jul 2024)

(Creates Robots with Advanced Mobility, Dexterity, and Intelligence including Spot, Atlas, and Stretch)

- Software Engineer (Mar 2022 - Jul 2024)
 - Led feature development of live telemetry display for LEL/hazardous gas and gamma radiation sensor data.
 - Developed map-based software applications that enhanced deployment efficiency and situational awareness.
 - Enhanced autonomous radiometric thermal inspection process that is used in all high-value engagements.
- Field Application Engineer (Mar 2020 - Mar 2022)
 - Supported customers in deploying Spot in nuclear, aerospace, and construction industries.
 - Developed software and hardware for radiation meters, GNSS, LiDAR, 360 cameras, and compute platforms.
 - Collected, reviewed, and delivered actionable feedback to improve Spot's abilities in various environments ranging from low-level locomotion to advanced autonomous behaviors.

SDSU Rocket Project, CA (Aug 2015 - Feb 2020)

(Student Team that Designs, Fabricates, and Launches Liquid and Solid Rockets)

- Senior Engineer/Advisor (May 2018 - Feb 2020)
 - Launched record-breaking liquid bi-propellant LOX/LCH4 rocket to an altitude of 13,205 feet.
 - Mentored and led new engineers in the development of avionics and propulsion systems.
 - Organized and analyzed cryogenic, static hot fire, and launch data leading to an increased understanding of system performance and design changes to improve launch success.
- President/Project Manager (May 2017 - May 2018)
 - Led a volunteer team of over 50 active members from multiple disciplines in hands-on engineering projects.
 - Built new systems and procedures that increased safety and improved documentation.
 - Managed one of the largest SDSU student organization budgets (\$250,000+) and secured funding through grant applications and private donors.

KegRocket, USA (Nov 2021 - Present)

(Hobbyists attempting to build a Liquid Rocket made out of Beer kegs because it's Fun)

- Actively developing software, onboard electrical systems, and ground support equipment. <https://www.kegrocket.com/>
- Primary console operator for pressurization tests, wet dress rehearsals, and cryogenic testing.